WORKED SOLUTION

STEP III 2017, Question 9

Mechanics • Particles connected by spring

STEP III 2017, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

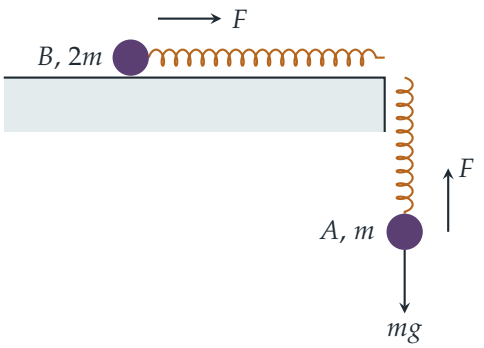
step.maths.org/questions/17-s3-q9

The key idea

The spring tension is unknown and changes with time, but it acts with equal size on both particles. So if we add the two equations of motion in the right proportions, the tension cancels and leaves something we can integrate straight away. In the second part, that integrated relation tells us where both particles are at time T and how their speeds are linked. Conservation of energy then gives a second equation, and the two together fix the speed of B.

Solution

We model A and B as particles and the spring as light. The table is smooth, and the spring runs smoothly over the edge of the table. Once released, A falls vertically and B slides along the table towards the edge. Let the tension in the spring be F at time t. While $x < a$, B is still on the table.



The relation $y + 2x = \frac{1}{2}gt^2$

Take y positive downwards for A and x positive towards the edge for B. By Newton's second law:

$$\begin{aligned} \text{for } A \text{ (vertically):} \quad & m\ddot{y} = mg - F, \\ \text{for } B \text{ (horizontally):} \quad & 2m\ddot{x} = F. \end{aligned}$$

(For B, the weight and the normal reaction cancel, and there's no friction.) Adding the two equations gets rid of F:

$$m\ddot{y} + 2m\ddot{x} = mg \quad \Longrightarrow \quad \ddot{y} + 2\ddot{x} = g.$$

Both particles start from rest with $x = y = 0$. Integrating once gives $\dot{y} + 2\dot{x} = gt$, with no constant because both velocities are zero at $t = 0$. Integrating again gives

$$y + 2x = \tfrac{1}{2}gt^2,$$

again with no constant, which is the required result.

> **Ask yourself** We don't know F, since it depends on the extension $y - x$, which keeps changing. How can we still say anything?

We never need F. It appears with opposite signs in the two equations once we multiply them by the right factors (here, just add them). The combination $y + 2x$ is $1/m$ times the momentum-like quantity $my + 2mx$, and the only external force driving it is A's weight.

The speed of B at time T

At time $T = \sqrt{6a/g}$, particle B reaches the edge, so $x = a$. Since $\frac{1}{2}gT^2 = \frac{1}{2}g \cdot \frac{6a}{g} = 3a$, the first part gives

$$y + 2a = 3a \quad \Longrightarrow \quad y = a.$$

So the extension of the spring at time T is $y - x = a - a = 0$. The spring is at its natural length again, and it stores no elastic energy.

Write u and v for the speeds of B and A at time T. The once-integrated relation $\dot{y} + 2\dot{x} = gt$ gives, at $t = T$,

$$v + 2u = gT = \sqrt{6ag}. \tag{1}$$

Energy. The table is smooth and the spring is light, so mechanical energy is conserved. At the start, both particles are at rest and the spring is unstretched. At time T, A has dropped a distance a and the spring is unstretched again, so the loss of potential energy mga has all become kinetic energy:

$$\tfrac{1}{2}mv^2 + \tfrac{1}{2}(2m)u^2 = mga \quad \Longrightarrow \quad v^2 + 2u^2 = 2ga. \tag{2}$$

Ask yourself Two equations, one of them quadratic: shouldn't we expect two possible speeds?

Substituting $v = \sqrt{6ag} - 2u$ from (1) into (2):

$$\begin{aligned} 6ag - 4\sqrt{6ag}\,u + 4u^2 + 2u^2 &= 2ga \\ 6u^2 - 4\sqrt{6ag}\,u + 4ag &= 0 \\ 3u^2 - 2\sqrt{6ag}\,u + 2ag &= 0. \end{aligned}$$

The discriminant is $4 \cdot 6ag - 4 \cdot 3 \cdot 2ag = 0$, so the quadratic is a perfect square and there's only one root:

$$(\sqrt{3}\,u - \sqrt{2ag})^2 = 0 \quad \Longrightarrow \quad u = \sqrt{\frac{2ag}{3}}.$$

So the speed of B when it reaches the edge is $\sqrt{2ag/3}$, as required. From (1), $v = \sqrt{6ag} - 2\sqrt{2ag/3} = \sqrt{2ag/3}$ as well. At that moment the two particles move with the same speed, which fits with the spring being at its natural length.

Checks

- The repeated root has a meaning. For fixed total 'momentum' $v + 2u$, the kinetic energy $\frac{1}{2}mv^2 + mu^2$ is smallest when $v = u$, and that smallest value is exactly mga here. So energy and (1) can only both hold if $u = v$, and then (1) gives $3u = \sqrt{6ag}$ directly.
- Solving the motion in full (which the question tells us not to do) gives $\omega T = 2\pi$, where $\omega^2 = 3\lambda/(2am)$: the extension goes through exactly one full oscillation and comes back to zero. A step-by-step numerical integration with $a = 1.3\,\mathrm{m}$ and $g = 9.8\,\mathrm{m\,s^{-2}}$ gives $x(T) = y(T) = a$ and both speeds equal to $2.9143\,\mathrm{m\,s^{-1}} = \sqrt{2ag/3}$.
- Dimensions: $\sqrt{ag}$ is a speed, and $\sqrt{a/g}$ is a time.

Common slips

- Forgetting that the spring has elastic energy in general. The argument works only because $x = y = a$ at time T, which makes the extension zero. Show this explicitly.
- Writing the energy equation alone and trying to solve it for u. It has two unknowns, so we need relation (1) as well.

Beyond STEP

The quantity $my + 2mx$ behaves like a momentum: internal forces (here the spring tension) cancel in it, and only external forces change it. That's the start of the mechanics

of systems of particles. In general you split the motion into a centre-of-mass part and a relative part, and here the relative coordinate $y - x$ performs simple harmonic motion. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas carefully.

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